AI Programming Foundations Project Summary

Author: Sukh Sandhu

Overview

I built a reusable data workflow around the scikit-learn Wine recognition dataset: https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html. The workflow ingests a CSV copy of the dataset, applies conservative cleaning functions, summarizes class-level chemistry patterns, and produces three visualizations that prepare the data for later AI modeling.

Dataset Description

The dataset contains 178 rows and 15 columns after adding a readable target-name field. Each row represents one wine sample, with numeric chemistry measurements such as alcohol, flavanoids, total phenols, and color intensity. The target variable identifies one of three cultivars. The class counts are {'class_1': 71, 'class_0': 59, 'class_2': 48}, which is balanced enough for exploratory work but still limited to the original benchmark sample.

Workflow Description

The notebook follows the required sequence: ingestion, cleaning, exploratory analysis, visualizations, and summary. Ingestion loads the CSV with pandas and displays the first rows. Cleaning standardizes column names, removes duplicate rows, and validates that chemistry measurements are nonnegative. The exploratory function groups observations by cultivar and reports key means. The visualization section creates a boxplot, scatter plot, and correlation heatmap.

Key Decisions and Assumptions

The cleaning design is intentionally transparent because reproducible data workflows should make transformation decisions easy to inspect and rerun (Danchev, 2022). I did not impute or remove valid observations because the benchmark table had no missing values in the executed notebook. The EDA focused on alcohol, phenols, flavanoids, and correlations because these variables show separability across cultivars and would influence future model design. The three figures were chosen to show a distribution comparison, a two-feature relationship, and possible multicollinearity.

Results and Interpretation

Figure 1 shows that alcohol distributions differ by cultivar, with class_0 having the highest mean alcohol value in this run. Figure 2 shows a visible relationship between total phenols and flavanoids, with cultivar clusters partially separated. Figure 3 highlights strongly correlated chemistry measurements; the strongest absolute correlations include ['total_phenols vs flavanoids', 'flavanoids vs total_phenols', 'od280_per_od315_of_diluted_wines vs flavanoids']. These patterns suggest that a later classifier may benefit from scaled numeric predictors while monitoring redundant features.

Responsible Practice (Bias and Data Quality)

Data handling could introduce bias if cleaning choices remove one cultivar more often than others or if measurement validation is applied without domain knowledge. The notebook therefore records row counts before and after cleaning and avoids aggressive transformations. Best practices for scientific computing emphasize version control, readable scripts, and repeatable execution, all of which reduce the chance that hidden workflow changes alter the analysis (Wilson et al., 2014).

Reproducibility

The folder includes `requirements.txt`, the CSV dataset, the executed notebook, saved figures, and local repository evidence with multiple commits and a development branch. Another reviewer can install dependencies, rerun `data_workflow.ipynb`, and compare the regenerated metrics and figures to the saved outputs.

References

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. Journal of Open Source Education, 5(56), 156. https://doi.org/10.21105/jose.00156

Wilson, G., Aruliah, D. A., Brown, C. T., Chue Hong, N. P., Davis, M., Guy, R. T., et al. (2014). Best Practices for Scientific Computing. PLoS Biology, 12(1), e1001745. https://doi.org/10.1371/journal.pbio.1001745

Scikit-learn developers. Wine recognition dataset documentation. https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_wine.html

Selected Figures

Figure 1. Alcohol distribution by cultivar.

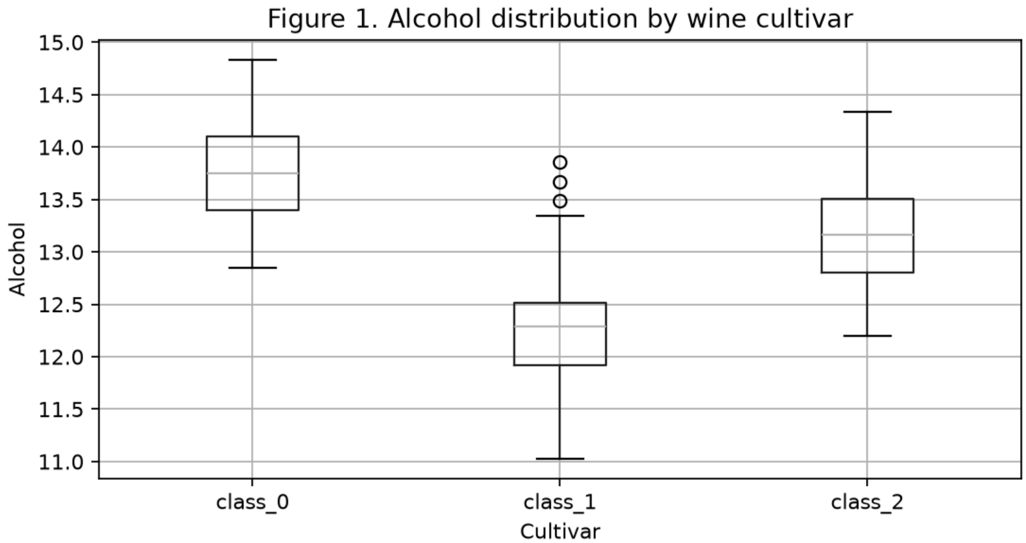

Figure 2. Phenols and flavanoids by cultivar.

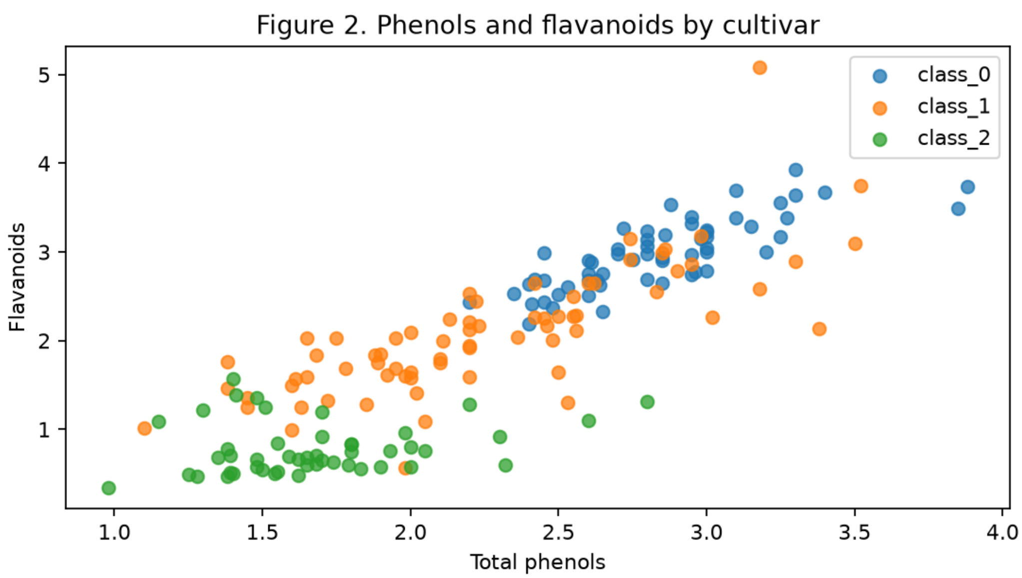

Figure 3. Feature correlation heatmap.

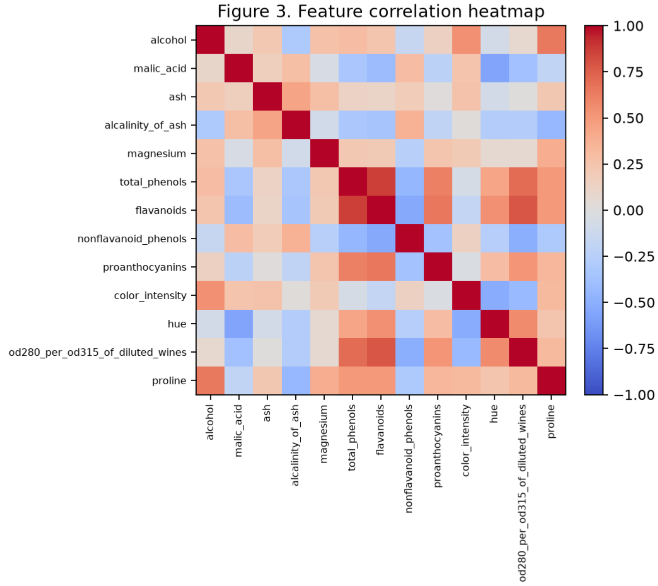

Figure 3. Feature correlation heatmap